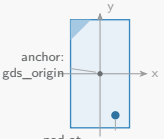
```
global = position + R(rotation.z) * M * local
```

die-local

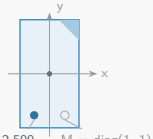


the die's own GDS frame, y-up

step 1, mirror

M = diag(-1, 1)

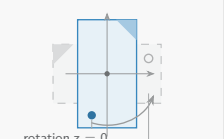
for orientation: flip_chip



only x changes sign; never mirror in y

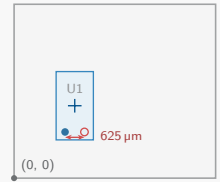
step 2, rotate

R(rotation.z)



step 3, translate by position

the canonical frame



a systematic offset of the whole pad field, not an obvious break; the GDS is well formed and the 3D view renders happily

the order matters. Rotating before mirroring gives a different answer for every angle that is not a multiple of 180 degrees

position does not move when the die rotates, which is why the contract puts it at the geometric centre rather than at a corner